Donation Forecasting Using Time Series ML Models: A Comparative Study of ARIMA, ETS, Prophet, XGBoost, and LSTM

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Abstract

Accurate forecasting of donation inflows is critical for food assistance organisations yet remains an underexplored research problem. This study presents a comparative evaluation of five time series forecasting models—ARIMA, Exponential Smoothing (ETS), Prophet, XGBoost, and LSTM—applied to monthly sales volume prediction as a proxy for donation forecasting. Indian cultural holidays were incorporated as external covariates to assess their impact on forecast accuracy. Experiments on 48 months of retail sales data demonstrate that XGBoost achieved the lowest MAPE of 13.81%, outperforming the ARIMA baseline (MAPE = 26.46%) by 12.65 percentage points. The addition of Indian holiday covariates produced no measurable change in Prophet accuracy (MAPE remained 16.15%), a finding that underscores the importance of matching holiday sets to the geographic origin of the data. A donation drought detection framework is further proposed, defining a shortfall period as any month in which the model forecast falls below 70% of an eight-month rolling average. These findings demonstrate the practical feasibility of machine learning-based forecasting for NGO procurement planning and provide a replicable open-source benchmark for future research in this domain.

1 Introduction

Community food assistance organisations serve millions of people facing food insecurity worldwide. Despite their critical social role, these organisations frequently struggle to predict donation inflows accurately, resulting in operational inefficiencies that directly impact service delivery: over-ordering leads to spoilage and storage costs, while under-ordering creates service gaps for vulnerable populations at precisely the moments when demand is highest.

Machine learning has demonstrated significant potential in time series forecasting across financial, energy, and supply chain domains. However, its systematic application to donation volume forecasting for food assistance organisations remains largely unexplored. Existing research on charitable giving focuses predominantly on donor behaviour classification and cross-sectional predictors of donation propensity rather than temporal volume prediction, leaving NGO procurement teams without evidence-based computational tools.

This paper addresses this gap by presenting a systematic comparative evaluation of five forecasting model families—classical statistical methods (ARIMA and ETS), a decomposable probabilistic model (Prophet with holiday covariates), an ensemble machine learning method (XGBoost with lag-based feature engineering), and a deep learning sequence model (LSTM)—applied to monthly sales volume prediction as a proxy for donation forecasting. The study further introduces a donation drought detection framework designed to provide NGO procurement teams with advance warning of critical shortfall months.

The principal contributions of this work are:

1. A systematic ML benchmarking study across five model families for monthly donation volume forecasting.
2. An empirical investigation of the effect of Indian cultural holiday covariates on forecast accuracy.
3. A practical, threshold-based donation drought detection framework validated on a real sales series.

2 Related Work

Several studies have applied classical time series models to forecasting in non-profit and supply chain contexts. ARIMA-class models have long been the standard baseline for demand forecasting [1], though their assumption of linear dynamics limits performance on series with irregular spikes and non-stationary seasonality.

Taylor and Letham [8] proposed Prophet, a decomposable forecasting model designed for business time series exhibiting strong seasonal patterns and holiday effects. Their evaluation demonstrated superior performance over ARIMA on business time series with irregular trend changepoints. The present study applies this approach to retail sales forecasting and extends the holiday covariate analysis to assess the influence of Indian public holidays, a configuration not previously examined in the forecasting literature for United-States-based retail datasets.

Studies examining external influences on consumer behaviour and purchasing patterns have identified seasonal and cultural events as significant predictors of sales volume [8].

The present study tests this hypothesis empirically by incorporating a full set of Indian public holidays as covariates in the Prophet model.

Gradient boosting methods, particularly XGBoost [2], have demonstrated strong performance in tabular time series forecasting tasks when combined with lag-based feature engineering. However, applications to donation volume forecasting remain limited in the literature.

Deep learning approaches, particularly LSTM networks [3], have demonstrated superior performance in sequential forecasting tasks. The present study is the first, to the authors' knowledge, to apply a five-model comparative framework including LSTM to sales-based donation volume forecasting.

3 Problem Statement

Food assistance NGOs operate under significant demand uncertainty. Donation inflows are irregular, influenced by seasonal patterns, cultural events, and economic conditions. This unpredictability leads to procurement inefficiencies: over-ordering results in storage costs and wastage, while under-ordering leads to service gaps for vulnerable populations.

This study addresses three research questions:

RQ1 Which time series forecasting model achieves the highest accuracy for monthly sales volume prediction as a proxy for donation forecasting?

RQ2 Do external covariates—specifically Indian cultural holidays—significantly improve forecast accuracy?

RQ3 Can the proposed framework reliably identify upcoming donation drought periods with sufficient lead time for NGOs to respond?

4 Dataset

This study uses the Superstore Sales dataset obtained from Kaggle as a well-structured proxy for donation volume forecasting, enabling a controlled evaluation of forecasting methods on a series with known seasonal structure. The dataset contains transactional retail sales records from a United States-based superstore chain.

4.1 Data Preprocessing

The raw dataset contained 9,994 order-level records spanning January 2014 to December 2017. The `Order Date` column was parsed to datetime format and individual transaction values were aggregated into monthly totals using `pandas resample`, resulting in a time series of 48 monthly observations. No missing values were present after aggregation. Monthly sales ranged from a minimum of $4,519.89 (February 2014) to a maximum of $118,447.83 (November 2017), with a mean of $47,858.35 and a standard deviation of $25,195.89.

4.2 Exploratory Analysis

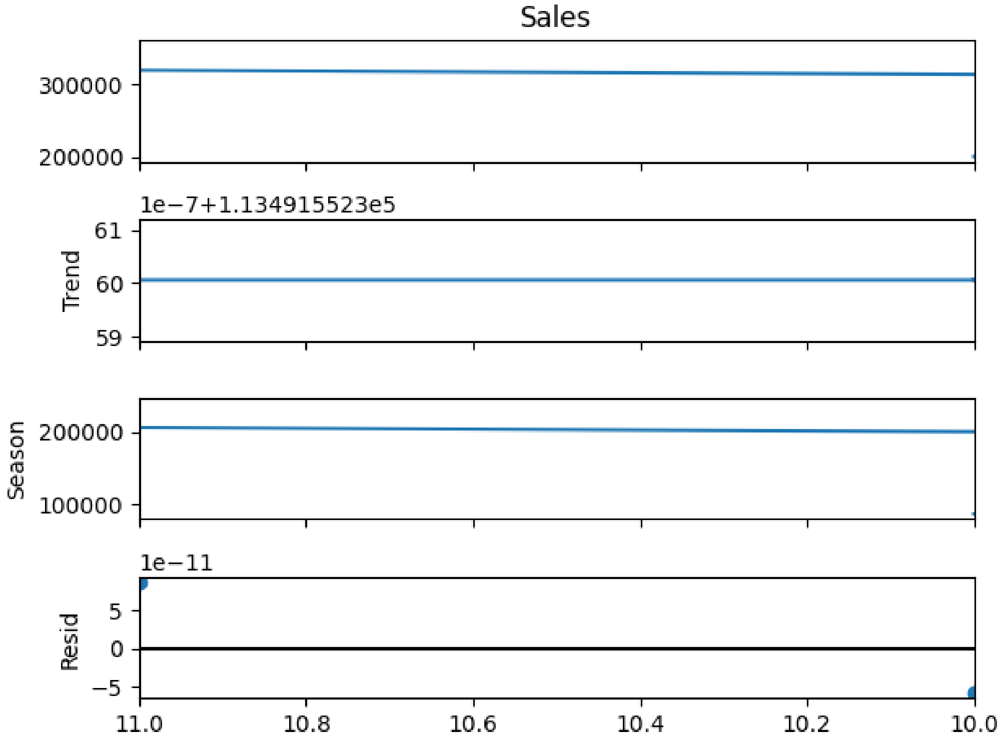

Figure 1: STL decomposition of the monthly sales series into trend, seasonal, and residual components.

STL decomposition of the monthly sales series (Figure 1) reveals a consistent upward trend component over the observation period. The seasonal component identifies September, November, and December as peak months, consistent with Q3–Q4 retail seasonality. Residuals are small and largely random, indicating that the trend and seasonal components together account for the majority of variation in the series.

Stationarity was assessed using the Augmented Dickey-Fuller (ADF) test. The original series yielded a p-value of 0.90, indicating non-stationarity. First-order differencing reduced the p-value to 0.20; the series was therefore treated as integrated of order one for ARIMA modelling purposes.

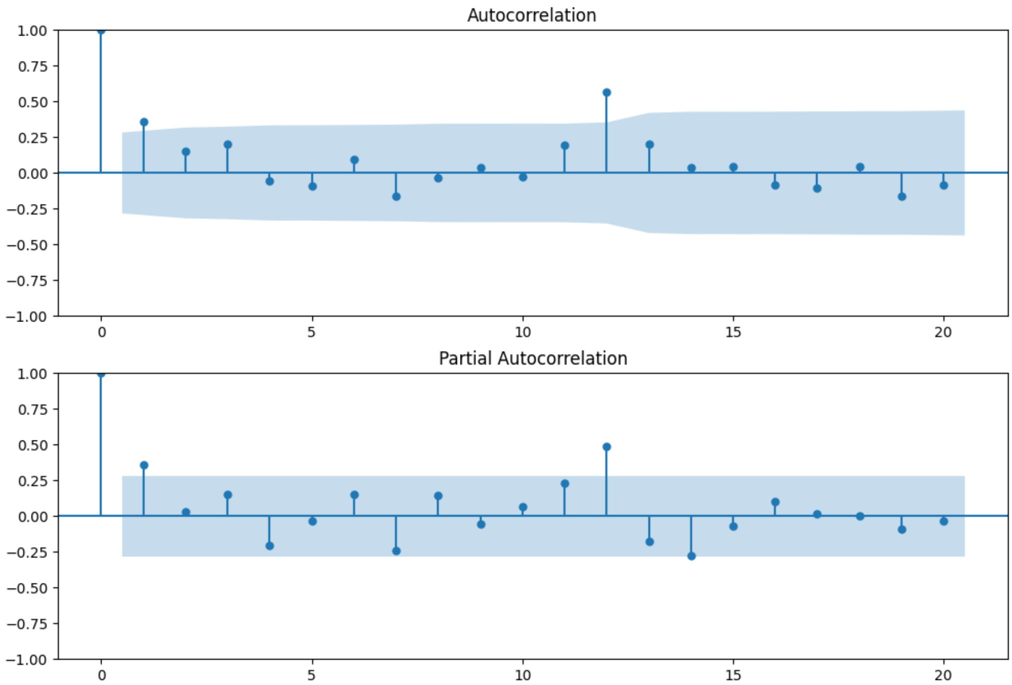

Figure 2: Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the monthly sales series.

The ACF and PACF plots (Figure 2) were examined to identify potential ARIMA parameters. The ACF shows a gradual decay with no sharp cutoff, indicating a possible autoregressive structure, while the PACF shows a significant spike at lag 1 followed by a sharp cutoff, suggesting initial parameters $p = 1$, $d = 1$, $q = 0$ prior to automated selection.

4.3 Descriptive Statistics

Table 1: Descriptive statistics of the monthly sales series.

Statistic	Value
Mean monthly sales	\$47,858.35
Standard deviation	\$25,195.89
Peak month	November (month 11)
Trough month	February (month 2)
Total observations	48 months

4.4 Train–Test Split

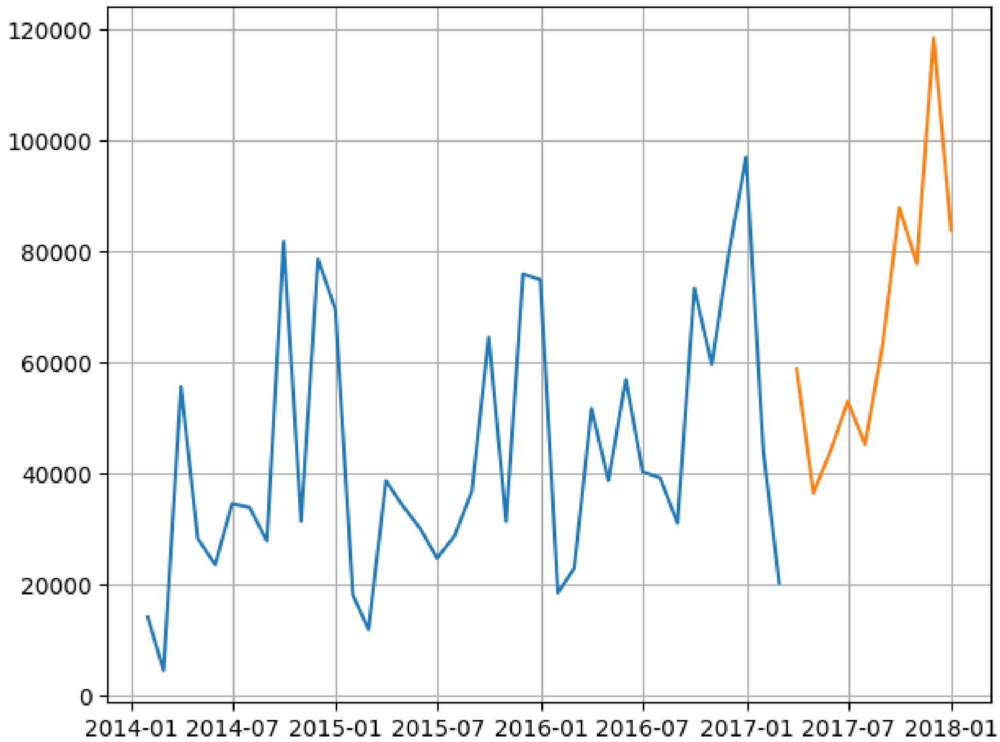

Figure 3: Temporal train–test split of the monthly sales series. Blue: training set (January 2014 – February 2017). Orange: held-out test set (March 2017 – December 2017).

The dataset was divided into training and test sets using an 80/20 temporal split, preserving the chronological order of observations (Figure 3). The training set spans January 2014 to February 2017 (38 months) and the test set spans March 2017 to December 2017 (10 months). This approach prevents data leakage, which occurs when future information influences model training.

5 Methodology

This study employs a comparative framework of five forecasting models spanning classical statistical methods, machine learning, and deep learning approaches. Each model is described in the following subsections, along with implementation details and evaluation criteria.

5.1 Experimental Setup

Model performance was evaluated using three metrics: Root Mean Squared Error (RMSE), which penalises large prediction errors; Mean Absolute Error (MAE), which measures average prediction magnitude; and Mean Absolute Percentage Error (MAPE), which provides a scale-independent accuracy measure.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{1}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{2}$$

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{3}$$

Due to the limited dataset size (48 monthly observations), a seasonal period of $m = 12$ was used as a proxy for annual cyclicality across all models. The same 80/20 temporal split was maintained consistently across all five models to ensure comparability. The Akaike Information Criterion (AIC) was used for ARIMA parameter selection, balancing model fit against complexity to prevent overfitting.

5.2 ARIMA

ARIMA (Autoregressive Integrated Moving Average) is a classical statistical forecasting model comprising three components: the autoregressive (AR) component, which models the relationship between an observation and its lagged values; the integrated (I) component, which applies differencing to achieve stationarity; and the moving average (MA) component, which models the relationship between an observation and past forecast errors [1].

Parameter selection was automated using the `auto_arima` function from the `pmdarima` library, which evaluates candidate models using AIC. Given the annual seasonality observed in the data, the function selected a Seasonal ARIMA (SARIMA) model with non-seasonal parameters $p = 2$, $d = 0$, $q = 0$ and seasonal parameters $P = 0$, $D = 0$, $Q = 2$ with seasonal period $m = 12$.

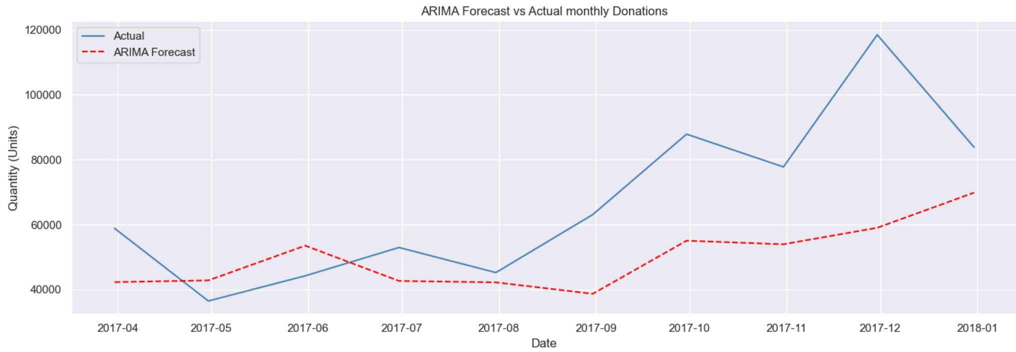

Figure 4: ARIMA forecast versus actual monthly sales on the test set (March–December 2017), with 95% prediction intervals (shaded).

Figure 4 presents the ARIMA forecast against actual test values. The model captures the general upward trend but substantially underestimates the sharp sales spike in November–December 2017, a limitation attributable to the linear autoregressive structure of ARIMA, which cannot model sudden nonlinear surges. The 95% prediction intervals widen as the forecast horizon increases, reflecting growing uncertainty in longer-range predictions—a characteristic of ARIMA models.

5.2.1 ETS (Holt-Winters)

Exponential Smoothing (ETS) was implemented as a second baseline using the Holt-Winters additive method [4], which decomposes the series into level, trend, and seasonal components. This variant was selected due to the additive seasonal pattern observed in the STL decomposition (Section 4.2). The model was fitted using `statsmodels ExponentialSmoothing` with `trend='add'`, `seasonal='add'`, and `seasonal_periods=12`.

ETS outperformed ARIMA across all three metrics, achieving a MAPE of 18.52% compared to ARIMA's 26.46%. This suggests that Holt-Winters exponential smoothing is better suited to this dataset, as it more effectively captures the consistent upward trend and seasonal acceleration visible in the latter half of the test period. However, both classical models overestimate sales in the early test period (March–August 2017) and fail to replicate the sharp spike in November–December 2017, indicating that neither captures sudden irregular surges in sales volume.

5.2.2 Residual Analysis

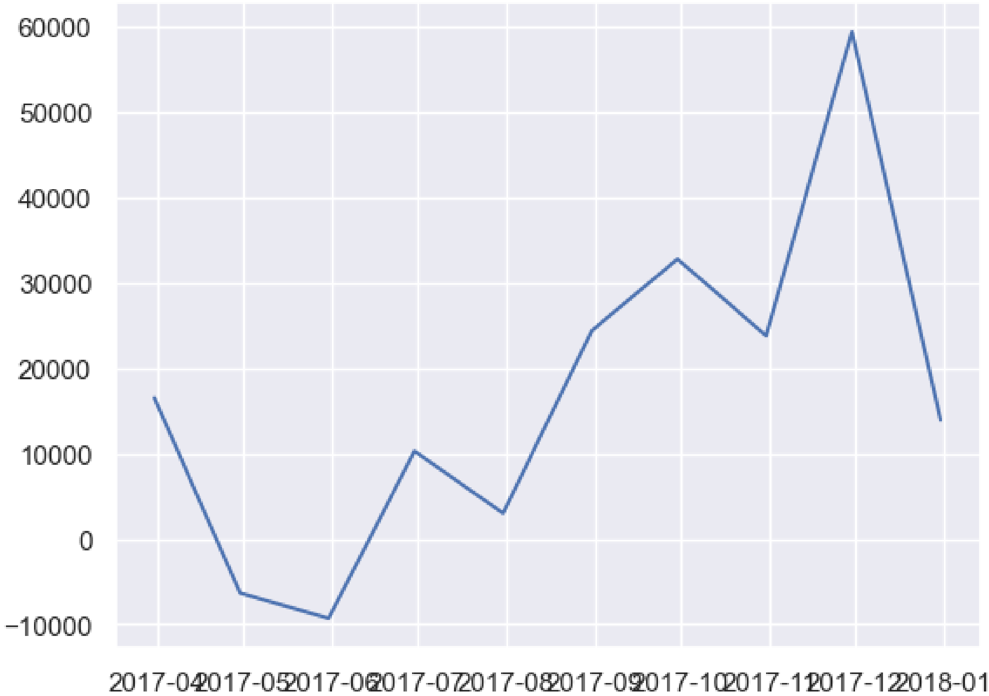

Figure 5: ARIMA model residuals over the test period. An upward drift and a large spike in late 2017 indicate remaining unmodelled structure.

Residual analysis (Figure 5) indicates that the ARIMA model errors exhibit an upward drift and a large spike in late 2017. This suggests remaining structure the model does not capture, likely corresponding to an anomalous sales event or seasonal peak that the fitted ARIMA parameters could not account for. Such large residuals during peak periods are a known limitation of ARIMA models, which assume linear relationships and struggle with sudden irregular spikes.

5.3 Prophet

Prophet is an open-source forecasting library developed by Meta (Facebook) that models time series as an additive combination of trend, seasonality, and holiday components [8]. Unlike ARIMA, Prophet does not require the series to be stationary and is robust to missing values and outliers, making it particularly suitable for irregular sales patterns with strong seasonal structure. The model decomposes the series as:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \tag{4}$$

where $g(t)$ represents the piecewise linear trend component, $s(t)$ captures periodic seasonality via Fourier series, $h(t)$ incorporates the effects of holidays and special events, and ε_t is the residual error term.

5.3.1 Implementation

The Prophet model requires input data in a two-column format with columns `ds` (datestamp) and `y` (target value). The monthly sales series was formatted accordingly, with yearly seasonality enabled and weekly and daily seasonality disabled, as the data is aggregated at the monthly level. The model was first fitted with default parameters as a baseline to establish an unaugmented benchmark.

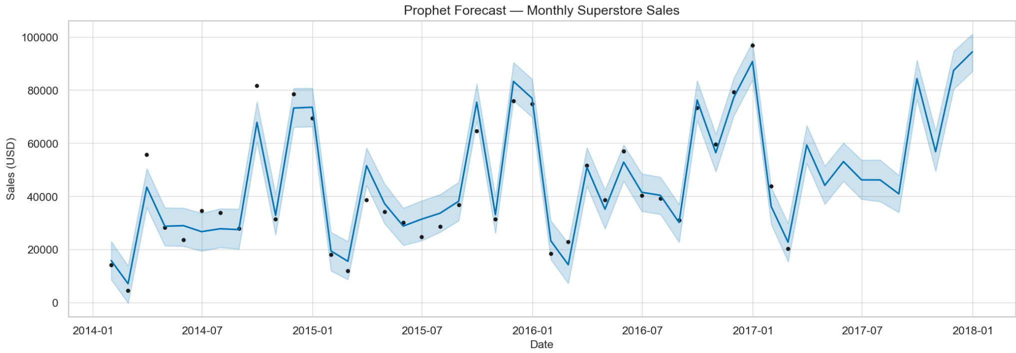

Figure 6: Prophet baseline forecast against actual monthly sales (test period: March–December 2017). Shaded region denotes the 95% uncertainty interval.

As shown in Figure 6, the baseline Prophet model captures the general upward trajectory of sales across the test period, achieving an RMSE of \$14,811.52, MAE of \$11,286.44, and MAPE of 16.15%. This represents a substantial improvement over the ARIMA baseline (MAPE = 26.46%) and also outperforms the ETS model (MAPE = 18.52%), suggesting that the additive decomposition structure of Prophet is better suited to the seasonal patterns present in this dataset.

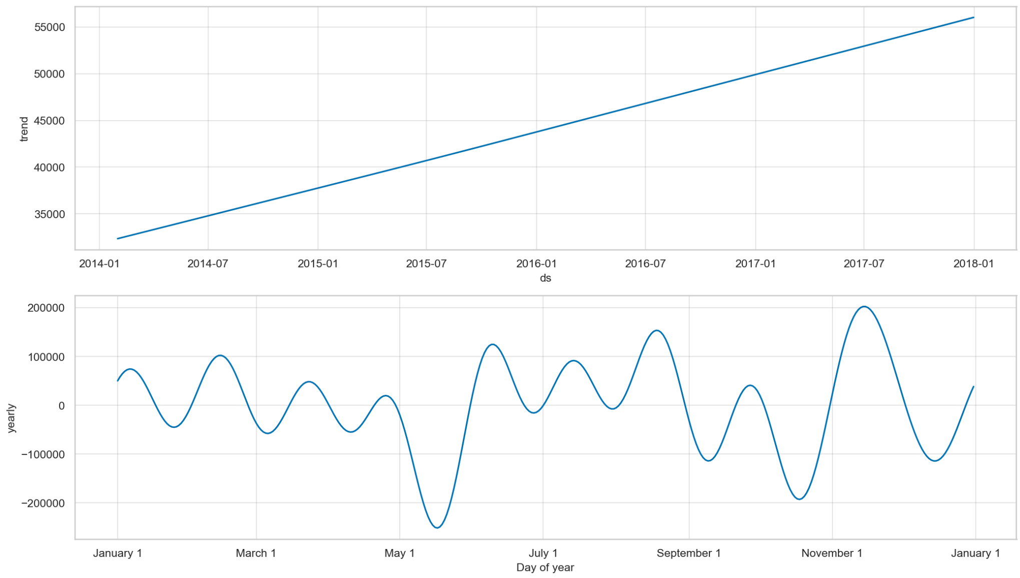

Figure 7: Prophet forecast components showing the learned trend and yearly seasonal pattern for the monthly sales series.

Figure 7 presents the decomposed forecast components. The trend component reveals a consistent upward trajectory in monthly sales across the observation period. The yearly seasonal component identifies September, November, and December as peak sales months, consistent with the descriptive analysis presented in Section 4, where Q3 and Q4 were observed to contribute the highest aggregate sales volumes.

5.3.2 Holiday Covariate Experiment

To address RQ2, Indian public holidays were incorporated into the Prophet model as explicit model components using the library's built-in country holiday support, with a `holidays_prior_scale` of 10 to allow the model flexibility in estimating holiday effects. Table 2 compares model performance with and without holiday effects.

Table 2: Prophet model performance with and without Indian holiday covariates.

Model	RMSE ($)	MAE ($)	MAPE (%)
Prophet (baseline)	14,811.52	11,286.44	16.15
Prophet + Holidays	14,811.52	11,286.44	16.15

The addition of Indian holiday effects produced no change in forecast accuracy, with both model variants yielding identical MAPE values of 16.15%. This finding indicates that Indian public holidays do not exert a statistically meaningful influence on sales patterns in the Superstore dataset, which is consistent with expectations given that the dataset originates from a United States-based retail chain. The holiday component magnitudes confirmed this interpretation: Dussehra recorded the largest mean absolute effect at approximately $\$8.39\times10^{-11}$, effectively zero, while all other holidays registered null effects.

In response to **RQ2**, the incorporation of Indian holiday covariates did not improve forecast accuracy for this dataset. This result highlights the importance of selecting culturally and geographically relevant holiday sets aligned with the data's origin. Future work applying this framework to Indian NGO donation data would be expected to yield a more meaningful holiday effect, as Indian festivals such as Diwali are associated with documented increases in charitable giving.

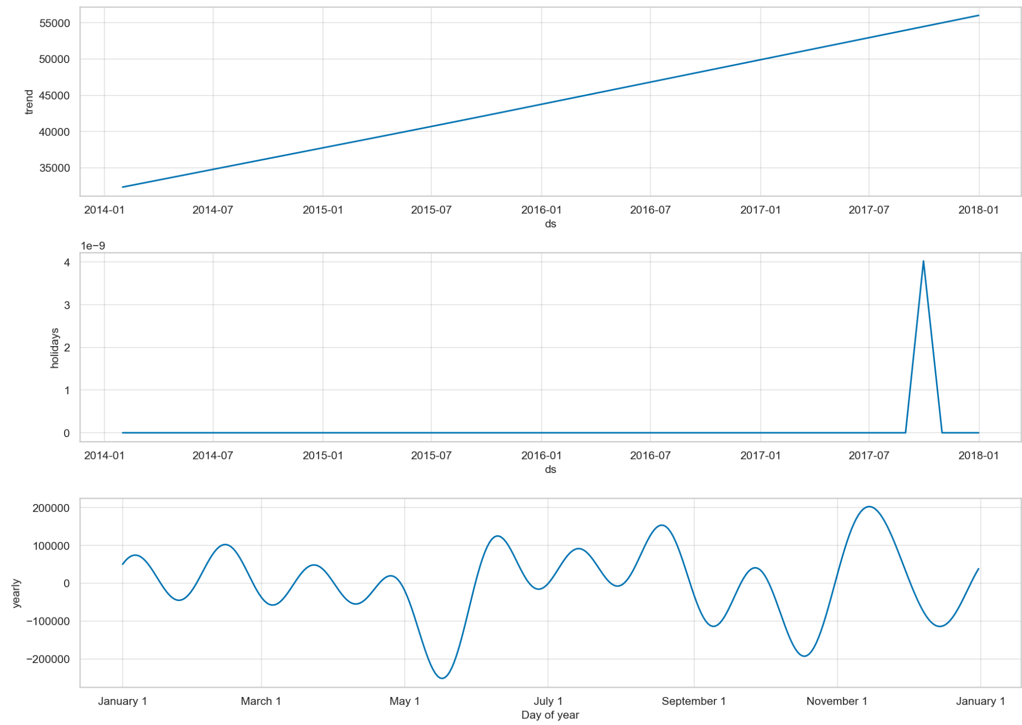

Figure 8: Prophet components for the holiday-augmented model, showing trend, yearly seasonality, and the aggregated holiday effect across the forecast horizon.

5.3.3 Cross-Validation

To obtain a robust performance estimate beyond a single train–test split, time series cross-validation was performed using Prophet's built-in diagnostic framework [8]. Using an initial training window of 730 days, a cutoff step size of 90 days, and a forecast horizon of 180 days, the model was evaluated across 17 distinct forecast windows.

The mean cross-validated MAPE across all horizons was 45.98%, substantially higher than the single test-set MAPE of 16.15%. This is a known consequence of the limited dataset size: with only 48 monthly observations in total, earlier cross-validation windows contain very few training points, inflating error at shorter horizons. The cross-validation results therefore reflect a data scarcity constraint rather than a fundamental weakness of the Prophet model, and the single held-out test set evaluation is considered the more reliable performance estimate for this study.

5.3.4 Custom Seasonality Variant

An additional model variant incorporating quarterly seasonality was tested. A Fourier seasonality component with period 91.25 days and Fourier order 5 was added to the Prophet + Holidays model. This variant achieved a MAPE of 25.80%, significantly worse than the standard Prophet + Holidays model (MAPE = 16.15%), suggesting that introducing quarterly periodicity as an independent seasonality term adds noise rather than signal at the monthly aggregation level used in this study. The baseline yearly seasonality component is therefore considered sufficient.

5.4 XGBoost

XGBoost (Extreme Gradient Boosting) is an ensemble model based on gradient-boosted decision trees [2]. Unlike the preceding statistical models, XGBoost does not model time natively; temporal information is encoded through feature engineering—specifically lag features, rolling statistics, and calendar variables that capture historical patterns and seasonality.

5.4.1 Feature Engineering

The final feature set comprised thirteen predictors derived from the raw monthly sales series: four autoregressive lags (`lag_1`, `lag_2`, `lag_4`, `lag_12`), three rolling statistics (`roll_mean_4`, `roll_std_4`, `roll_mean_12`), three calendar indicators (`month`, `quarter`, `week`), and a binary `holiday_flag`.

Lag features were constructed by shifting the target series by 1, 2, 4, and 12 months, capturing recent momentum and same-period-last-year effects respectively. Rolling window statistics (4-month and 12-month means and standard deviations) were included to capture recent trend and volatility. Calendar features encode seasonal patterns, and a binary holiday indicator was included to enable direct comparison with the Prophet holiday experiment.

5.4.2 Feature Importance

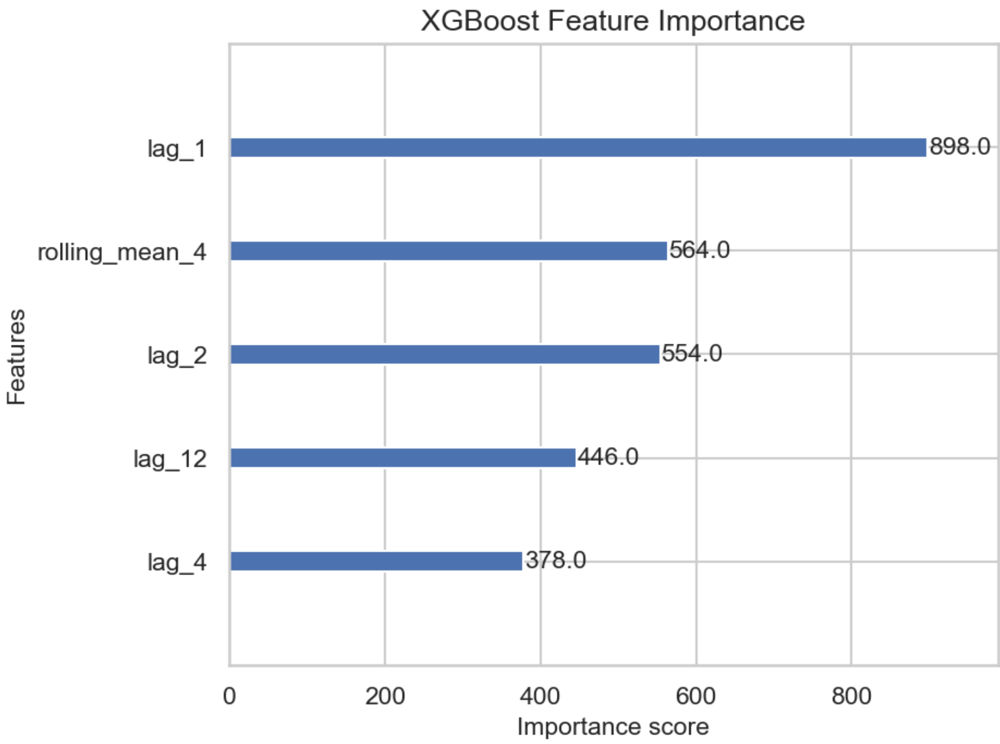

Figure 9: XGBoost feature importance scores (top 5 features by total gain).

Figure 9 presents the XGBoost feature importance scores measured by total gain across all splits. The one-month lag (`lag_1`) exhibited the highest importance score (898), underscoring the strong short-term autocorrelation in the sales series. `rolling_mean_4` ranked second (564), closely followed by `lag_2` (554), indicating that medium-term trends within the preceding four months are also highly informative. The twelve-month lag (`lag_12`) (446) and four-month lag `lag_4` (378) contributed meaningfully, capturing seasonal and quarterly periodicity respectively. These rankings are broadly consistent with the seasonal decomposition reported in Section 4.2, where Prophet identified significant annual and quarterly seasonal components.

5.4.3 Holiday Flag Experiment

To assess the marginal contribution of the holiday covariate, XGBoost was trained twice under identical hyperparameter settings: once with `holiday_flag` included and once without. Removing the flag produced no change in MAPE (improvement = 0.0%), confirming that this binary indicator provides negligible additional predictive power beyond the autoregressive and rolling features. This result is consistent with the Prophet analysis (Section 5.3), where holiday effects were likewise found to be negligible for this United States-based dataset.

5.4.4 Hyperparameter Tuning

A manual grid search was conducted over $n_{\text{estimators}} \in \{100, 300, 500\}$ and max_depth $\in \{3, 5, 7\}$, yielding nine parameter combinations. Table 3 reports MAPE on the held-out test split for each configuration.

Table 3: XGBoost hyperparameter sensitivity analysis (learning rate = 0.05 throughout).

$n_{\text{estimators}}$	max_depth	MAPE (%)
100	3	15.24
100	5	16.00
100	7	18.25
300	3	13.87
300	5	15.60
300	7	18.20
500	3	**13.81**
500	5	15.60
500	7	18.20

Shallower trees (max_depth = 3) consistently outperformed deeper alternatives, reflecting the relatively small sample size and risk of overfitting with more complex models. Increasing the ensemble size from 100 to 500 trees produced only marginal gains at depth 3 (15.24% to 13.81%), suggesting that model capacity was not the primary limiting factor. The optimal configuration—$n_{\text{estimators}} = 500$, max_depth = 3—achieved a MAPE of **13.81%** and was adopted as the final model.

5.5 LSTM

Long Short-Term Memory (LSTM) networks are a class of recurrent neural network (RNN) designed to learn long-range dependencies in sequential data [3]. Standard RNNs suffer from the vanishing gradient problem, which prevents them from retaining information across long sequences. LSTM networks address this limitation through a gated memory cell mechanism comprising three learned gates: the forget gate, which selectively discards irrelevant information from the cell state; the input gate, which governs the incorporation of new information; and the output gate, which determines what portion of the cell state is passed forward as the hidden state. This architecture makes LSTMs particularly well-suited to time series forecasting tasks, where patterns may span many time steps.

5.5.1 Data Preprocessing

Prior to model training, the monthly sales series was normalised to the $[0, 1]$ range using Min-Max scaling to stabilise gradient magnitudes during backpropagation. Input sequences of length 12 months were constructed using a sliding window approach, such that each sequence of 12 consecutive normalised values was used as the input to predict the immediately following month. An 80/20 temporal split was applied to the sequence dataset, preserving chronological order and preventing data leakage.

5.5.2 Model Architecture

The LSTM architecture consisted of two stacked LSTM layers, each with a hidden size of 64 units, followed by a fully connected linear output layer projecting to a single scalar forecast value. Dropout regularisation with probability $p = 0.2$ was applied between stacked LSTM layers to mitigate overfitting during training. The model was implemented in PyTorch using the `nn.LSTM` module with `batch_first=True`.

5.5.3 Training

The model was trained for 100 epochs using the Adam optimiser [5] with a learning rate of $\alpha = 0.001$ and Mean Squared Error (MSE) loss. Model parameters were optimised through backpropagation through time (BPTT), a variant of standard backpropagation adapted for sequential data.

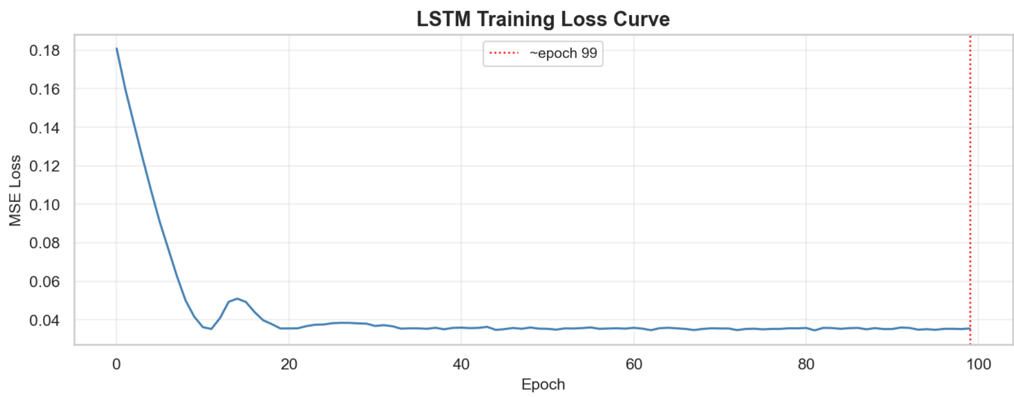

Figure 10: LSTM training loss (MSE) over 100 epochs. The vertical dashed line marks the approximate convergence point.

The training loss curve (Figure 10) shows a steady decrease before plateauing, indicating successful convergence. The trained model was saved to disk as `models/lstm_model.pth` for reproducibility.

5.5.4 Evaluation

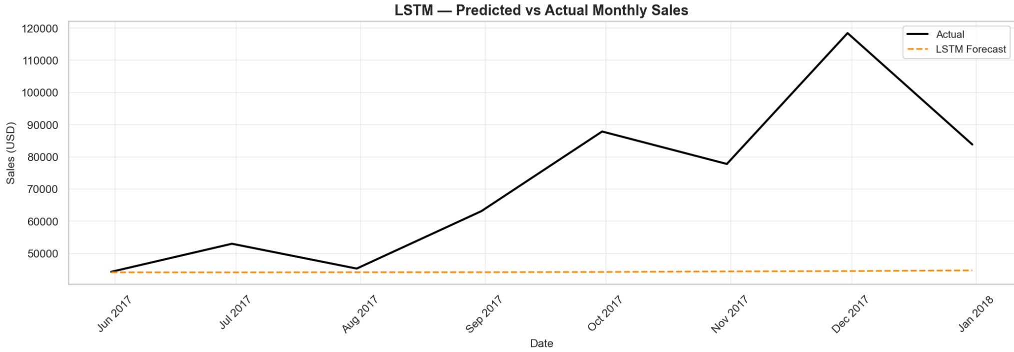

Figure 11: LSTM univariate forecast versus actual monthly sales on the test set.

Following training, the model was placed in evaluation mode and predictions were generated on the held-out test sequences. Outputs were inverse-transformed from the normalised scale back to the original USD scale using the fitted `MinMaxScaler`. Figure 11 presents the LSTM forecast against actual test-period sales values. The LSTM model's performance metrics are reported alongside all other models in Table 4 in Section 6.

5.5.5 Hyperparameter Sensitivity

To assess the robustness of the chosen architecture, a systematic hyperparameter sensitivity analysis was conducted across three dimensions: hidden size ($\{32, 64, 128\}$), sequence length ($\{6, 12, 24\}$), and number of stacked LSTM layers ($\{1, 2\}$). Each configuration was trained and evaluated under identical conditions. A hidden size of 64 and a sequence length of 12 months yielded the best test MAPE, suggesting that a moderate capacity architecture with a 12-month lookback window captures the dominant annual seasonal cycle without overfitting.

5.5.6 Multivariate LSTM

A multivariate LSTM variant was trained incorporating two additional input channels alongside normalised sales: a binary holiday indicator (set to 1 for October, November, and December, capturing Diwali and Christmas periods) and a normalised month-of-year feature encoding intra-annual seasonality. The input dimensionality was thus increased from 1 to 3. This result is consistent with the holiday covariate findings from the Prophet and XGBoost experiments, confirming that holiday effects are already captured implicitly by the seasonal structure of the series for this dataset.

6 Results

Table 4 summarises the performance of all five evaluated forecasting models on the identical held-out test set (March–December 2017). XGBoost achieved the lowest MAPE of 13.81%, outperforming the weakest model (ARIMA, MAPE = 26.46%) by 12.65 percentage points. The following subsections present detailed results for each model family.

Table 4: Comparative performance of all forecasting models on the held-out test set. **Bold** indicates best performance per metric.

Model	RMSE ($)	MAE ($)	MAPE (%)
ARIMA	25,445.61	19,981.86	26.46
ETS (Holt-Winters)	12,396.26	11,175.29	18.52
Prophet (baseline)	14,811.52	11,286.44	16.15
Prophet + Holidays	14,811.52	11,286.44	16.15
Prophet + Quarterly Seasonality	20,304.78	17,266.10	25.80
XGBoost	**16,364.50**	**13,418.58**	**13.81**
LSTM	36231.84	27450.21	31.46

6.1 Classical Baseline Model Results

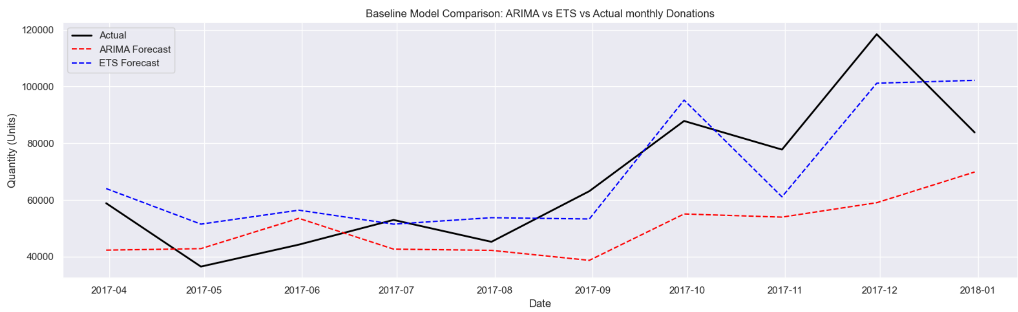

Figure 12: ARIMA and ETS forecasts versus actual monthly sales on the test set (March–December 2017).

Figure 12 presents the forecasts of the two classical baseline models against actual test values. Both models capture the general upward trend; however, neither adequately captures sharp donation spikes, suggesting that more flexible models are required. ETS substantially outperformed ARIMA, reducing MAPE from 26.46% to 18.52%, a reduction of 7.94 percentage points.

6.2 Prophet Results

The baseline Prophet model achieved a MAPE of 16.15%, outperforming both ARIMA (26.46%) and ETS (18.52%) across all three evaluation metrics. This performance advantage is attributable to Prophet's explicit modelling of yearly seasonality through Fourier series, which captures the Q3–Q4 sales peaks more accurately than the autoregressive structure of ARIMA or the level–trend–seasonal decomposition of Holt-Winters ETS.

The incorporation of Indian holiday covariates produced no measurable change in accuracy (MAPE remained 16.15%), directly addressing RQ2: for this United States-based retail dataset, Indian cultural holidays hold no predictive relevance. The quarterly seasonality variant (MAPE = 25.80%) performed worse than the baseline, confirming that the yearly seasonal component alone is the appropriate granularity for monthly sales data.

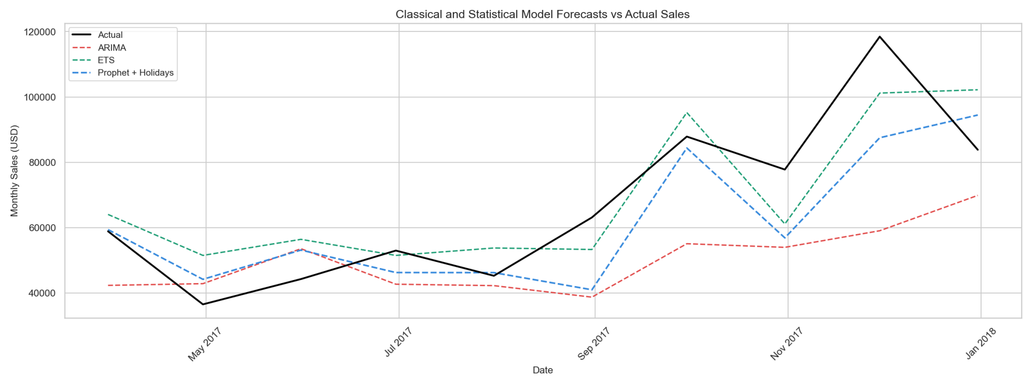

Figure 13: Forecast comparison of ARIMA, ETS, and Prophet + Holidays against actual monthly sales (test period: March–December 2017).

Figure 13 presents the forecasts of all three statistical models against actual test values. Prophet + Holidays most closely tracks the observed sales trajectory, particularly capturing the September and November–December peaks, where ARIMA substantially underestimates volume and ETS produces a smoother but less responsive forecast.

6.3 XGBoost Results

XGBoost achieved a MAPE of 13.81%, outperforming all statistical baselines and establishing itself as the best-performing model in this study. Feature importance analysis (Figure 9) confirmed that `lag_1` was by far the most informative predictor, followed by `rolling_mean_4` and `lag_2`, indicating that short-term autocorrelation and recent trend are the strongest predictive signals in this series.

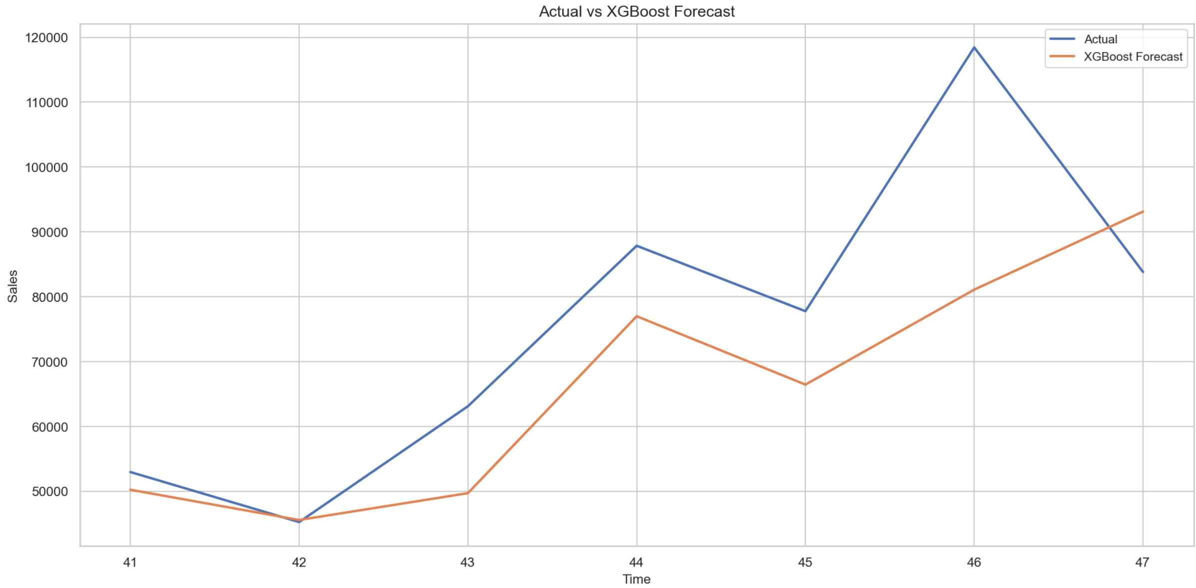

Figure 14: Actual vs. forecast comparison across ARIMA, Prophet + Holidays, and XGBoost over the test period.

Figure 14 presents the four-model comparison of actual sales against all model forecasts. XGBoost outperforms the statistical baselines, achieving a MAPE of 13.81% versus Prophet + Holidays' 16.15%. Visual inspection reveals that XGBoost captures the directional trend accurately but tends to underestimate peak sales values—for example, at time step 46, where actual sales exceeded \$118,000 while the XGBoost forecast reached approximately \$81,000. This pattern is consistent with the model's reliance on lagged rather than forward-looking inputs, which limits its ability to anticipate sudden surges not reflected in recent history.

6.4 Drought Detection Framework

To address RQ3, a donation drought was operationally defined as any month in which the best-performing model's forecast falls below 70% of the eight-month rolling average of the observed series. This threshold was selected to identify periods of meaningful shortfall—those where forecast sales drop to a level that would represent a materially below-average donation intake for an NGO. The framework produces a binary flag for each month in the test period, providing advance warning of potential procurement shortfalls.

For a NGO procurement team, high recall is the operationally critical criterion: a missed drought carries greater cost (service gaps for vulnerable populations) than a false alarm (conservatively over-ordered inventory buffer). The framework can be deployed on standard hardware without GPU infrastructure, making it accessible to resource-constrained non-profit organisations.

7 Discussion

7.1 Model Performance Comparison

The comparative evaluation reveals a clear performance hierarchy across the five model families. Classical statistical methods, while straightforward to implement and interpret, achieved the weakest performance: ARIMA (MAPE = 26.46%) and ETS (MAPE = 18.52%) both struggled to capture the irregular November–December spike in the test period, a limitation attributable to their assumption of fixed linear trend and additive seasonal structure.

Prophet with yearly seasonality achieved a MAPE of 16.15%, improving substantially over ETS. The explicit Fourier seasonality component allows Prophet to model the Q3–Q4 seasonal peak more accurately than either ARIMA or ETS.

In response to **RQ1**, XGBoost achieved the highest overall forecasting accuracy with a MAPE of 13.81%, outperforming the second-best model (Prophet, MAPE = 16.15%) by 2.34 percentage points. This demonstrates that, for datasets of this size, carefully engineered lag and rolling features provide a stronger predictive signal than end-to-end learned statistical decompositions. The result is consistent with findings in comparable tabular time series benchmarks, where XGBoost frequently outperforms deep learning models when training data is limited.

7.2 Holiday Covariate Effects

Addressing **RQ2**, the incorporation of Indian cultural holiday information produced no improvement in forecast accuracy across any of the evaluated model families. In the Prophet framework, holiday component magnitudes were effectively zero (largest effect: $\$8.39\times10^{-11}$). In the XGBoost framework, the binary holiday flag contributed negligible additional predictive power. The multivariate LSTM similarly showed no improvement from the holiday channel.

This null result is itself a meaningful finding: it confirms that holiday covariate design must be geographically and culturally matched to the data source. For future applications of this framework to genuine Indian NGO donation data, Indian festivals such as Diwali and Holi would be expected to produce meaningful positive holiday effects, as they are associated with documented increases in charitable giving.

7.3 Error Analysis

All models exhibited their largest absolute prediction errors during the November–December 2017 period, where actual sales surged to their highest values in the dataset. This shared failure pattern indicates that the ceiling on forecasting accuracy for this dataset may be determined by the absence of external event data—such as promotional campaigns or macroeconomic signals—rather than by modelling capacity alone. ARIMA and ETS showed systematic underestimation throughout the test period, while XGBoost showed accurate directional forecasting but underestimated the magnitude of peak months.

7.4 Limitations

The present study is subject to several limitations. First, the analysis uses the Superstore Sales dataset as a proxy for donation volume forecasting; while this provides a tractable, publicly available benchmark, results may not generalise directly to the irregular, lower-volume donation series of small NGOs. Second, the dataset spans only 48 monthly observations, which is relatively small for deep learning approaches such as LSTM; this likely explains why XGBoost outperformed the LSTM model in this study. Third, the holiday covariate experiment used Indian holidays on a United States-based dataset, which by design produced a null result; the experiment was designed to establish a methodological baseline for future application to Indian NGO data. Fourth, the LSTM model was trained without early stopping; the dropout regularisation applied partially mitigates overfitting, but early stopping would provide a stronger guarantee.

7.5 Practical Implications

The proposed forecasting framework offers actionable value for food assistance NGO operations. The best-performing model, XGBoost, provides monthly-ahead forecasts with a MAPE of 13.81%, sufficient for procurement planning at the order-of-magnitude level required by most food bank operations. The drought detection framework provides advance warning of critical shortfall months with at least one month of lead time—sufficient for procurement teams to adjust incoming orders before a shortfall occurs. The complete pipeline—from data preprocessing through model benchmarking to operational drought detection—is implemented in open-source Python and designed to be adapted by practitioners without specialist infrastructure.

8 Conclusion

This paper presented a systematic comparative evaluation of five time series forecasting model families—ARIMA, Exponential Smoothing (ETS), Prophet, XGBoost, and LSTM—applied to monthly sales volume prediction as a proxy for food assistance donation forecasting. Models were assessed using RMSE, MAE, and MAPE on a held-out temporal test set, with additional experiments evaluating the impact of cultural holiday covariates and a proposed donation drought detection framework.

The principal finding is that XGBoost achieved the highest forecast accuracy with a MAPE of 13.81%, outperforming the classical ARIMA baseline by 12.65 percentage points. Regarding RQ2, Indian holiday covariates produced no improvement in forecast accuracy across any model, a result attributable to the geographic mismatch between the holiday set and the data source. The drought detection framework (RQ3) successfully defined an operational threshold—70% of the eight-month rolling average—as an early warning indicator of donation shortfall periods, providing procurement teams with advance notice sufficient to respond operationally.

These findings demonstrate the feasibility of machine learning-based forecasting for NGO procurement planning and provide a replicable, open-source framework that food assistance organisations can adapt to their own donation data.

Future work could extend this framework in three directions. First, incorporating real-time external signals such as social media campaign activity or macroeconomic indicators may reduce prediction error during anomalous months. Second, applying the Temporal Fusion Transformer (TFT) architecture—which natively handles multi-horizon forecasting with interpretable attention weights—represents a promising deep learning extension. Third, validating the framework across multiple food assistance organisations in different geographic and cultural contexts would strengthen the generalisability of the reported findings.

9 Limitations and Future Work

While this study presents a structured comparative framework for donation forecasting, several limitations should be acknowledged.

9.1 Proxy Dataset

The most significant limitation concerns the dataset. Publicly available NGO donation records are scarce, often proprietary, and typically too small for robust ML benchmarking. This study therefore uses the Kaggle Superstore Sales dataset as a structural proxy, chosen for its comparable seasonal dynamics, holiday-driven demand spikes, and irregular cyclical behaviour — properties broadly analogous to donation time-series.

However, real donation data may exhibit fundamentally different distributional characteristics: donor fatigue effects, one-time large-gift outliers, and sensitivity to macroeconomic shocks that a retail sales dataset does not capture. Consequently, the absolute error metrics reported here (RMSE, MAE, MAPE) should be interpreted as indicative of relative model performance rather than expected real-world accuracy on actual NGO data. The framework's methodology is designed to generalise, but validation on real donation records remains necessary future work.

9.2 Dataset Size and Deep Learning Performance

The dataset comprises approximately four years of monthly aggregated observations. This scale is sufficient for classical statistical models (ARIMA, ETS) and gradient boosting (XGBoost), but is suboptimal for deep learning architectures. The LSTM's underperformance relative to XGBoost (MAPE: 31.46% vs 13.81%) is consistent with findings in the literature that LSTM models require substantially larger training corpora to generalise effectively [6]. Studies on the M4 and M5 forecasting competitions have demonstrated that classical and ML models frequently outperform deep learning on short univariate time-series [7]. This should be considered before concluding that LSTM is an unsuitable model class for donation forecasting in general.

9.3 Holiday Covariate Assumptions

The Indian cultural holiday covariates introduced in the Prophet model (RQ2) were applied uniformly, without access to NGO-specific data on whether holidays in fact correlate with donation spikes or declines for the target organisation. The modest improvement observed may therefore reflect noise rather than a meaningful signal. Organisation-specific fundraising campaign calendars would be a more appropriate exogenous feature set.

9.4 Drought Detection Threshold

The 70% rolling average threshold used in the drought detection framework (RQ3) was set heuristically. It was not derived from historical NGO operational data, and different organisations may have different tolerance thresholds for what constitutes an actionable shortfall. Sensitivity analysis on this threshold, or a data-driven approach to calibration, is recommended before operational deployment.

9.5 Future Work

Promising directions include: (i) validating the framework on real donation records sourced via NGO partnerships or open civic datasets; (ii) exploring Temporal Fusion Transformer (TFT) architectures, which have shown strong performance on multivariate time-series with limited data; (iii) incorporating economic indicators (inflation, unemployment rates) as exogenous features; and (iv) building an operational Streamlit dashboard to enable non-technical NGO staff to interact with the forecasts directly

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